ME2 Computing- Coursework summary

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A) What physics are you trying to model and analyse? (Describe clearly, in words, what physical phenomenon you wish to analyse)

The objective is to model the transport phenomena of lithium ions within the electrolyte of a lithium-ion battery during discharge. More specifically, this report will analyse how the concentration of lithium ions throughout the electrolyte changes over time as they move from the anode to the cathode under the influence of concentration gradients (diffusion) and electric fields (convection).

Diffusion is a fundamental transport process in which molecules move from regions of higher concentration to regions of lower concentration, aiming to reach equilibrium. Convection in this context considers the effect of the potential difference between the two electrodes which drives the electrons in the external circuit (the basic mechanism behind how batteries provide current) and affects the lithium ions within the electrolyte due to their positive charge.

The model takes inspiration from an iPhone battery (electrolyte ~50 µm in width, discharge time ~5-6 hours). Since discharging is considered, the initial concentration of lithium ions is highest at the anode, decreasing exponentially along the electrolyte. As time passes, the concentration of ions at the anode is assumed to exponentially decay. On the other hand, at the cathode we assume all the lithium ions are being consumed in the chemical reaction.

B) What PDE are you trying to solve, associated with the Physics described in A? (write the PDE)

$$\frac{\partial c(x,t)}{\partial t} = D\frac{\partial^2 c}{\partial x^2} - \mu E\frac{\partial c}{\partial x} \qquad \text{Eq. 1}$$

Where c is the concentration of lithium ions, D the diffusion coefficient, μ the mobility of lithium ions, E the electric field, t the time passed since discharge began, and x the length along the electrolyte ($x = 0$ at the anode, $x = 50\mu\text{m}$ at the cathode).

C) Boundary value and/or initial values for my specific problem: (be CONSISTENT with what you wrote in A)

The initial condition (equation 2) models an exponential decay through space of the concentration, with a maximum of c_h mol/m^3, a minimum of c_l mol/m^3, and a decay constant of α (µm^{-1}).

$$c(x,0) = c_l + (c_h - c_l)e^{-\alpha x} \qquad \text{Eq. 2}$$

The same exponential decay model (but with decay constant k in s^{-1}) models the concentration of ions at the anode through a Dirichlet boundary condition:

$$c(0,t) = c_l + (c_h - c_l)e^{-kt} \qquad \text{Eq. 3}$$

At the cathode, the 100% reaction efficiency can be reflected by a zero flux Neumann condition:

$$\left.\frac{\partial c}{\partial x}\right|_{x=50\mu\mathrm{m}} = 0 \qquad \text{Eq. 4}$$

D) What numerical method are you going to deploy and why? (Describe, in words, which method you intend to apply and why you have chosen it as opposed to other alternatives)

Explicit method – Forward Time, Central Space (FCTS)

The FTCS is a straightforward method which discretizes the time derivative using a forward difference and the space derivative using a centred difference. When the stability conditions are met, the FCTS method is computationally efficient. The following are the conditions to be met for stability:

$$\frac{(\mu E)\Delta x}{D} \leq 2 \qquad \text{(convective stability)}$$

$$\frac{D\Delta t}{(\Delta x)^2} \leq \frac{1}{2} \qquad \text{(diffusive stability)}$$

The method is first-order in time and second-order in space due to the space and timestep constraints.

Alternative explicit methods, such as the Lax-Wendroff method and the Upwind Scheme, were considered. The former, despite its higher, second order time accuracy, may result in oscillations near sharp gradients. The latter is better tailored to handle advection problems by considering also flow direction, so the FCTS is preferable when dealing with 1-D convection-diffusion problems.

Implicit method – Crank-Nicolson

The Crank-Nicolson method is an implicit finite difference technique which 'averages the explicit forwards Euler scheme and the implicit backward Euler scheme. It is a second-order accurate method in both time and space while presenting unconditional stability for linear diffusion problems, providing flexibility on parameter values and grid size. Despite these advantages with respect to the FCTS method, Crank-Nicolson solutions are prone to decaying spurious oscillations[1]. Additionally, it is more computationally intensive than any explicit method, as it requires a linear system of equations to be solved at each time step.

Alternatives include the Backward Euler method and implicit Runge-Kutta methods. The former, despite offering stronger stability and immunity to oscillations, is only first order accurate in time, presenting no benefit over the explicit FCTS method. The latter, although flexible and offering high-order accuracy, involves solving more complex system of non-linear equations, significantly increasing the computational burden.

[1] https://www.ma.imperial.ac.uk/~ajacquie/IC_Num_Methods/IC_Num_Methods_Docs/Literature/DuffyCN.pdf

E) I am going to discretise my PDE as the following (show the steps from continuous to discrete equation and boundary/initial conditions:

Explicit method

The following steps are taken to discretise the PDE using the FCTS method:

$$\frac{\partial c(x,t)}{\partial t} = D\frac{\partial^2 c}{\partial x^2} - \mu E\frac{\partial c}{\partial x}$$

$$\frac{\partial c}{\partial t} \Longrightarrow \frac{c^{p+1} - c^p}{\Delta t}$$

$$\frac{\partial^2 c}{\partial x^2} \Longrightarrow \frac{c_{i+1} - 2c_i + c_{i-1}}{(\Delta x)^2}$$

$$\frac{\partial c}{\partial x} \Longrightarrow \frac{c_{i+1} - c_{i-1}}{2\Delta x}$$

$$\frac{c_i^{p+1} - c_i^p}{\Delta t} = D\frac{c_{i+1}^p - 2c_i^p + c_{i-1}^p}{(\Delta x)^2} - \mu E\frac{c_{i+1}^p - c_{i-1}^p}{2\Delta x} \qquad \text{Eq. 4}$$

Rearranging to solve for c_i^{p+1}:

$$c_i^{p+1} = \left(D\frac{c_{i+1}^p - 2c_i^p + c_{i-1}^p}{(\Delta x)^2} - \mu E\frac{c_{i+1}^p - c_{i-1}^p}{2\Delta x}\right)\Delta t + c_i^p \qquad \text{Eq. 5}$$

Out of the initial and boundary conditions, only the cathode boundary condition involves a derivative and must be discretized. It must be noted that a backwards finite difference is used since $i+1$ is outside the space domain:

$$\left.\frac{\partial c}{\partial x}\right|_{x=L} = 0 \Longrightarrow \frac{c_i^p - c_{i-1}^p}{\Delta x} = 0 \Longrightarrow c_i^p = c_{i-1}^p \qquad \text{Eq. 6}$$

Implicit method

Equation 7. shows the discretization of the PDE according to the Crank-Nicolson method:

$$\frac{c_i^{p+1} - c_i^p}{\Delta t} = \frac{D}{2}\left(\frac{c_{i+1}^{p+1} - 2c_i^{p+1} + c_{i-1}^{p+1}}{(\Delta x)^2} + \frac{c_{i+1}^p - 2c_i^p + c_{i-1}^p}{(\Delta x)^2}\right) - \frac{\mu E}{2}\left(\frac{c_{i+1}^{p+1} - c_{i-1}^{p+1}}{2\Delta x} + \frac{c_{i+1}^p - c_{i-1}^p}{2\Delta x}\right) \qquad \text{Eq. 7}$$

Rearranging to isolate the c^{p+1} terms on the LHS and the c^p terms on the RHS:

$$\frac{c_i^{p+1}}{\Delta t} - \frac{D}{2}\left(\frac{c_{i+1}^{p+1} - 2c_i^{p+1} + c_{i-1}^{p+1}}{(\Delta x)^2}\right) + \frac{\mu E}{2}\left(\frac{c_{i+1}^{p+1} - c_{i-1}^{p+1}}{2\Delta x}\right) = \frac{c_i^p}{\Delta t} + \frac{D}{2}\left(\frac{c_{i+1}^p - 2c_i^p + c_{i-1}^p}{(\Delta x)^2}\right) - \frac{\mu E}{2}\left(\frac{c_{i+1}^p - c_{i-1}^p}{2\Delta x}\right)$$

Isolating each c term to build the matrix and vector used to solve the linear system of equations:

$$\left(-\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right)c_{i+1}^{p+1}+\left(\frac{1}{\Delta t}+\frac{D}{(\Delta x)^2}\right)c_i^{p+1}-\left(\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right)c_{i-1}^{p+1}$$
$$=\left(\frac{D}{2(\Delta x)^2}-\frac{\mu E}{4\Delta x}\right)c_{i+1}^{p}+\left(\frac{1}{\Delta t}-\frac{D}{(\Delta x)^2}\right)c_i^{p}+\left(\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right)c_{i-1}^{p}$$

With these coefficients we can set up a linear system of equations represented in equation 8:

$$\mathbf{Ac}=\mathbf{b} \qquad \text{Eq. 8}$$

Where $\mathbf{A}$ is an $N\times N$ matrix, with N being the number of nodes in the space discretization, $\mathbf{c}$ is the ion concentration vector at the next timestep, and $\mathbf{b}$ is a vector with N rows. $\mathbf{A}$, $\mathbf{c}$, and $\mathbf{b}$ are displayed below:

LHS:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \dots \\ \left(-\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right) & \left(\frac{1}{2\Delta t}+\frac{D}{(\Delta x)^2}\right) & -\left(\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right) & 0 & 0 & \dots \\ 0 & \left(-\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right) & \left(\frac{1}{2\Delta t}+\frac{D}{(\Delta x)^2}\right) & -\left(\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right) & 0 & \dots \\ \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \\ \dots & 0 & 0 & \left(-\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right) & \left(\frac{1}{2\Delta t}+\frac{D}{(\Delta x)^2}\right) & -\left(\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right) \\ \dots & 0 & 0 & 0 & \frac{1}{\Delta x} & \frac{-1}{\Delta x} \end{pmatrix} \begin{pmatrix} c_0^{p+1} \\ c_1^{p+1} \\ c_2^{p+1} \\ \vdots \\ c_{N-1}^{p+1} \\ c_N^{p+1} \end{pmatrix}$$

RHS:

$$\begin{pmatrix} c_0^0\times e^{-kt} \\ \left(\frac{D}{2(\Delta x)^2}-\frac{\mu E}{4\Delta x}\right)c_2^p+\left(\frac{1}{\Delta t}+\frac{D}{(\Delta x)^2}\right)c_1^p+\left(\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right)c_o^p \\ \left(\frac{D}{2(\Delta x)^2}-\frac{\mu E}{4\Delta x}\right)c_2^p+\left(\frac{1}{\Delta t}+\frac{D}{(\Delta x)^2}\right)c_1^p+\left(\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right)c_o^p \\ \vdots \\ \left(\frac{D}{2(\Delta x)^2}-\frac{\mu E}{4\Delta x}\right)c_N^p+\left(\frac{1}{\Delta t}+\frac{D}{(\Delta x)^2}\right)c_{N-1}^p+\left(\frac{D}{2(\Delta x)^2}+\frac{\mu E}{4\Delta x}\right)c_{N-2}^p \\ 0 \end{pmatrix}$$

Solving this linear system of equations premultiplying by $\mathbf{A}^{-1}$ for every timestep yields the solution array c.

F) Plot the numerical results comprehensively and discuss them (discuss how the results describe the physics and comment on any discrepancies or unexpected behaviours). Use multiple types of visual graphs. Present and discuss any outcomes of the grid analysis, as requested in Task 9, too.

Variation in time

Figure 1 displays the variation in time of the concentration of lithium ions at different points in the electrolyte.

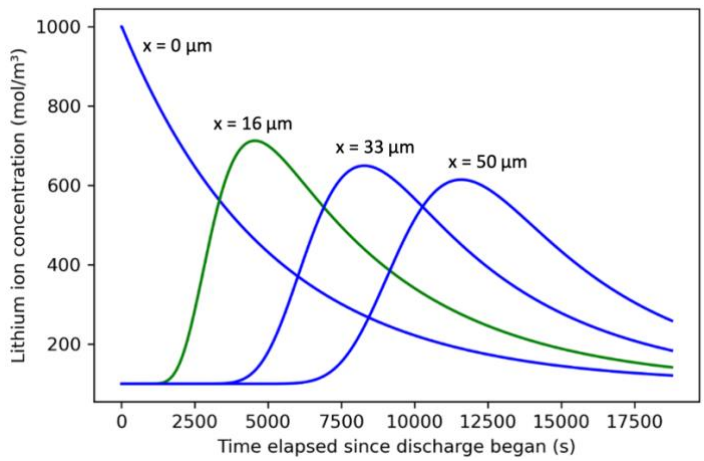

Figure 1. Concentration profile over time at different locations

Variation in space

Figure 2 displays the variation in space of the concentration of lithium ions at different times.

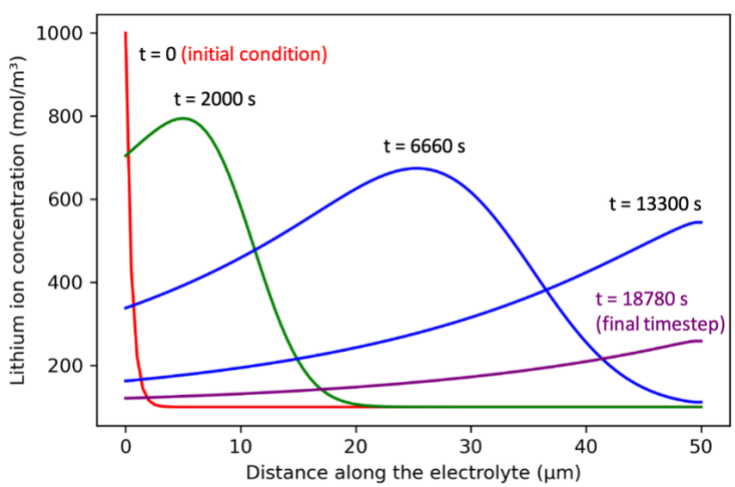

Figure 2. Concentration profile across the electrolyte at different points in time

NOTE: The results in Figures 1 and 2 obtained explicitly and implicitly are indistinguishable, hence are shown once.

Animations

The following animation presents a heat map visualization of the concentration profile.

The following animation compares the explicit and implicit methods.

NOTE: To speed up the animations to last around 20 seconds, one in every ten frames is shown since the minimum framerate allowed by the Matplotlib animation library has already been reached.

The figures and animations show a general migration of ions from the anode to the cathode, which agrees with the physical phenomena described in section A. The battery is considered fully discharged when the average concentration across the electrolyte is less than 170 mol/m^3. This yields a discharge time of 18,780 s or 5 hours and 13 minutes, consistent with the current standard for iPhone batteries.

<u>Grid analysis</u>

For the explicit method, halving the space step ($\Delta x = 0.25$) causes instability and convergence issues, as the diffusive stability term increases to more than 0.5. This can be seen in Figure 3(a) and (b), which plot the concentration profile at the same location or time as the green lines in Figures 1 and 2.

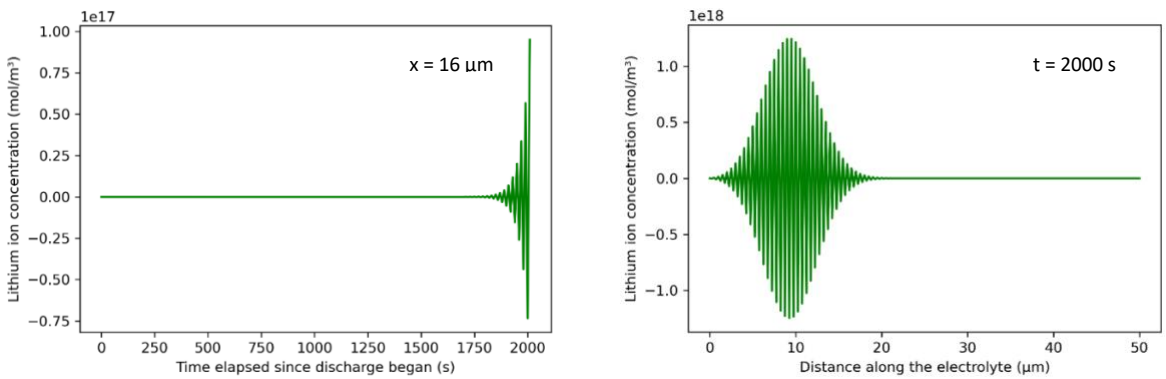

Figure 3. Concentration profile over (a) time and (b) space with a finer grid

On the other hand, doubling the space step ($\Delta x = 1$) decreases granularity and hence accuracy of the solution. This can be seen in Figure 4, again plotting the concentration profile at the same location or time.

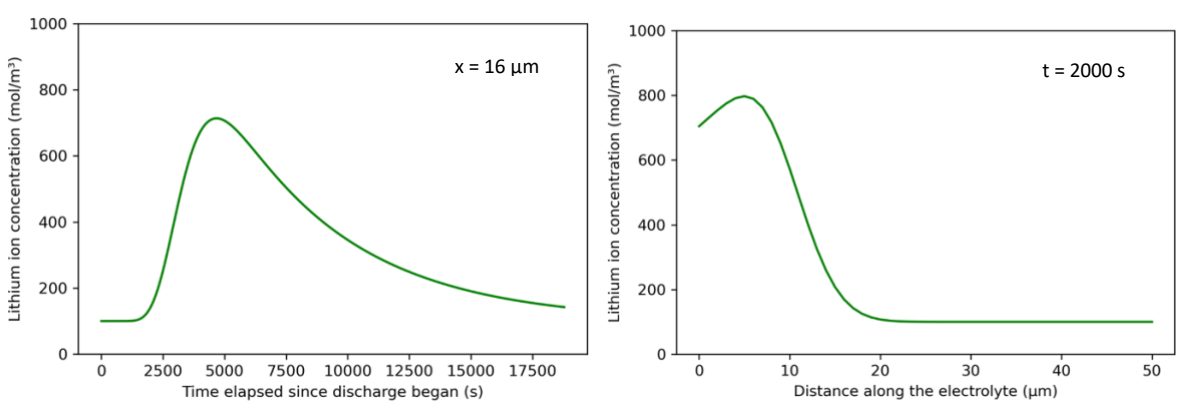

Figure 4. Concentration profile over (a) time and (b) space with a coarser grid

Applying a finer grid to the implicit method does not affect the results other than increasing computation time, granularity, and hence accuracy; applying a coarser grid has the same effect as it does for the explicit method.

G) Other remarks (limits of the model, convergence problems, possible alternative approaches, anything you find relevant and important to mention):

The model employs several assumptions and simplifications that cause it to stray away from reality. Factors such as the rate of reaction and intercalation of the lithium ions at the electrodes, temperature, and current affect the flux of lithium ions as well, and D and E are not truly constants. Moreover, the boundary conditions are simplistic, and the point of full discharge is not physically accurate. In short, the model can get arbitrarily complex. However, as is the case for all models, there must be a compromise between complexity and computational efficiency and accuracy. As long as the shortfalls of the model are appreciated, and the results are validated by testing, this is not an issue.

In terms of computation time, both the implicit and the explicit methods with the base parameters executed quickly (no longer than 10 seconds). However, decreasing the step size in the space domain increased computation time significantly for the implicit method. This is due to the increase in size of the matrix and the inefficient matrix inversion which had to be performed at each time step. A possible solution is to implement a tridiagonal matrix algorithm, which efficiently computes systems of linear equations for tridiagonal matrices like the one used in this case.

Overall, provided the stability conditions are met, the explicit and implicit methods agree and provide effective ways of solving the PDE modelling the flux of lithium ions through the electrolyte of a battery.

NOTE

If the hyperlinks to the animations do not work, try:

https://imperiallondon-my.sharepoint.com/personal/jl5722_ic_ac_uk/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Fjl5722%5Fic%5Fac%5Fuk%2FDocuments%2FYear%202%2FComputing%20Coursework%2Frealone%2Egif&parent=%2Fpersonal%2Fjl5722%5Fic%5Fac%5Fuk%2FDocuments%2FYear%202%2FComputing%20Coursework&ga=1

https://imperiallondon-my.sharepoint.com/personal/jl5722_ic_ac_uk/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Fjl5722%5Fic%5Fac%5Fuk%2FDocuments%2FYear%202%2FComputing%20Coursework%2Fotherrealone%2Egif&parent=%2Fpersonal%2Fjl5722%5Fic%5Fac%5Fuk%2FDocuments%2FYear%202%2FComputing%20Coursework&ga=1